R37-02 kinematic image: conditional result for P1, P4 and P6

Owner: HTT inference / obsstat feature extraction. Claim tier: C0 conditional methods. Transfer source: none. The Gaia E1/B1 and R36 E2/STF5 laws are separate; their covariance is unknown. No native solver, observational geometric vorticity, matter shear, or original-42 admission is inferred. Original acceptance remains 17/42 (analytic alternate 18/42). Full signed tensors, matrices and source hashes are in `KINEMATIC_IMAGE_RESULT.json`.

Same-input Gaia E1/B1 comparison

The saved 24-vector is three reported-redshift-bin E1 polar/B1 axial six-vectors followed by six within-bin $g(z) = z/(1+z)$ slopes. It is a 755,850-row producer's Gaussian *working summary*, not a new catalogue fit or spatial-covariance validation. Profiling the 18 arbitrary bin intercepts leaves the original six signed slope coordinates $(6.60158047, 0.29169945, -1.50131566, -10.18496521, -7.64665391, 0.48156053)$ microarcsec/year and exactly the saved 6×6 covariance. Their scalar Mahalanobis power is retained in the JSON. The $24 \to 18$ bin-centre map loses all six of these slopes under free-bin nuisance; it must not be interpreted as zero physical depth slope.

A *different,* explicitly constrained bin family assumes one common six-vector intercept and either g or g^2 depth response. On the same saved 18×18 covariance, the signed g slope is $(0.78422940, -1.57176426, 0.56325278, 21.94228237, -10.02218535, -6.28569937)$ microarcsec/year. The three isolated-bin scalar powers are 48.19295, 59.98241, 52.96719; those powers alone lose signs and are not an independent joint test. The previous saved 100,000-draw matched working-law experiment gives joint GLS/tensorized coverage 0.94927 and power 0.42130 at the stated Mahalanobis 2.5 injection. Diagonal-bin nominal errors give coverage 1.00000 and power 0; the joint-sandwich repair gives coverage 0.94976 and power 0.40135. These are Monte Carlo diagnostics under the producer's fixed Gaussian law; `COMPRESSION_COMPARISON.json` retains MCSE, exact trial counts, power MCSE and binomial intervals. `information_comparison.csv` is a compact rate table.

For g versus g^2, the nuisance-profiled union response has rank 12. Full-data and 12D-common-carrier profiled deviance differences are equal to 5.7×10^{-14} in this run (both approximately -67.77514657); a six-coordinate score optimized only for g gives zero difference.

The discarded-space adequacy difference is therefore the full -67.77514657 for this particular comparison. An invertible tensor-coordinate change retains the same GLS information. A scalar energy test has exactly the same omnibus statistic as joint GLS, while a signed predeclared-direction test can be more powerful for its different target. Neither the common carrier nor five-factor bookkeeping creates information beyond correct joint GLS.

Separate PM history and nullspace

R36-03 supplies an independent E2/STF5 first-order selected-design Gram law. In the fixed common-distance image let $q = (B_o - B_s)/T_{ref}$ and $s = \dot{B}_o$, both in the saved microarcsec/year coefficient units, with dimensionless endpoint tensors B_o, B_s and $k = 2cT_{ref}/D$. The five identified rate coefficients are $u = kq - s$; $k = 0$ is the distant fixed-endpoint limit. After spin/glide-six nuisance the five information eigenvalues are $(2.43693, 2.65563, 3.17145, 3.67145, 4.95102)$ in inverse squared rate units. For each supplied $k \in \{0, 0.1, 1, 2, 10\}$, the ten-coordinate (q, s) inverse fibre is $\{(q, s) : s = kq - u\}$: five unbounded null directions $\delta s = k\delta q$. If separate B_o, B_s coordinates are introduced, their common shift adds five more null directions. A free E2 nuisance removes all five identified coefficients.

For a **stipulated sensitivity**, write the fitted rate vector $a = u + e$ with coefficient-level STF5 Frobenius error $\|e\|_F \leq r$ microarcsec/year. This is an assumed body, not a measured physical remainder bound. Reusing the primary saved selected-sample five-coefficient estimate $\hat{a}$, Fisher matrix F, and $\chi^2_5(0.95) = 11.070497693516351$, the identified quotient image is $E_{95}(\hat{a}, F) + B_5(r)$. Its directional interval is $v^T\hat{a} \pm [\sqrt{11.070497693516351\, v^T F^{-1} v} + r\|v\|]$, and an eigenmode halfwidth is $\sqrt{11.070497693516351/\lambda_i} + r$, entirely in microarcsec/year. At $r = 0$ the usual conditional Gaussian ellipsoid remains; each unit of assumed rate remainder adds one unit to each unit-eigenvector halfwidth. `physical_region_vs_remainder.pdf` plots this support, while `KINEMATIC_IMAGE_RESULT.json` retains the centre, basis, widths and full unbounded fibre. The original R36 power-80 best/worst radii remain 1.60963/2.29431 microarcsec/year under its fixed design. Unknown catalogue frame spin and centroid-to-matter/history response preclude geometric-vorticity and matter-history claims.

Two optical bridge paragraphs for the current P6 supplement

The qualified R32-05 optical inverse and R36-04 joint response can be composed on the retained selected-source mathematical control. In one observer rest frame with signature $(-+++)$, outward direction, $H > 0$, $d = -A/(cH)$ and $S = \sigma/H$, the signed magnitude octupole obeys $O = \kappa L_S d + R_3$, $\kappa = 5/\ln 10$, and $L_S d = \mathrm{STF}(d \otimes S)$. Here d, S are dimensionless and the 96-direction saved fit provides ten signed direct-dipole/octupole coordinates with their complete 10×10 covariance. The P6 apparent tensor has $\|S\|_F = 0.03265680189321492$; using it as a response coefficient does not establish a common emitting congruence. A physical acceleration or shear image additionally needs the same-congruence source law, finite-distance and selection bounds, and an H normalization. The seven-component perpendicular residual lies in a four-dimensional out-of-response subspace for fixed nonzero S; at $S = 0$ all seven directions are perpendicular. It cannot be assigned to a kinematic parameter.

For this *newly stipulated* synthetic Gaussian control, a single 95% ten-coordinate ellipsoid supplies outer direct and octupole radii of the same event. R36 supplied the covariance and deterministic response, not a calibrated probability law; the 95% interpretation holds only under this added Gaussian assumption. Profiling the direct radius once over $\mathcal{D} = \{\|d - d_{obs}\| \le r_d, \|d\| \le 0.03\}$ and using $\epsilon = 0.08$ gives one STF3 residual budget $B = E_3 + e_O + B_z + B_{sel} + |\kappa|\sqrt{3/5} e_S D_{max} = 0.00187216048709$ mag with $B_z = B_{sel} = e_S = 0$ only for this declared mathematical control. The profiled residual is 9.194×10^{-5} mag, so this control is compatible; the earlier R36 unit-budget counterexample remains incompatible. Direct uncertainty is not added again to B. At fixed nonzero S, the inverse-image ellipsoid is centred at $\hat{d} = (\kappa L_S)^+ O$; its simple outer radius is $\sqrt{\max(B^2 - \|O_\perp\|^2, 0)}/s_{\min}(\kappa L_S)$, intersected with both direct and physical balls. This radius exceeds the declared physical cap in the baseline control, so the control does not tightly identify d. The full direct/octupole covariance cross-block has Frobenius norm $1.7820478375 \times 10^{-12}$ in mixed units, and the complete residual covariance is retained. The plotted extra-error axis is an assumed sensitivity, not a measured CF4 error. At $S = 0$ the inverse is undefined and only $\|O\| \le B$ plus direct-region nonemptiness is tested; the owned consumer distinguishes exact zero by maximum absolute entry, so $10^{-8}S$ and representable $10^{-200}S$ retain the scaled inverse; it explicitly rejects nonrepresentable inverse arithmetic. This is a numerical-domain statement, not a physical identification.

Fixed-Q exact sign-radius supplement

For a fixed simple-spectrum STF2 tensor Q, a nonzero chirality $\chi(Q, O)$, and an ambient STF3 Frobenius error ball in O, let $v = C_Q O$ and $M_Q = \|Q\|_F^2 I/3 + 2Q^2/5$. In the oriented eigenframe of Q, the exact distance to zero chirality is $\rho_{sign} = \min_i |v_i|/\sqrt{(M_Q)_{ii}}$: the zero set is the union of three hyperplanes and orthogonal projection attains each distance. Reusing the executed sharp witness $Q = \mathrm{diag}(-1/\sqrt{2}, 0, 1/\sqrt{2})$ gives $\rho_{sign} = 1/\sqrt{3} = 0.5773502691896256$ and an attaining STF3 perturbation, checked numerically. This fixed-Q result does not cover uncertain Q, a fixed-norm O sphere, or a physical parity inference without a qualified source response.

Figures `optical_joint_compatibility.pdf`, `physical_region_vs_remainder.pdf`, `GLS_tensor_scalar_comparison.pdf`, and `chirality_radius.pdf` are vector PDFs. Their source/status/allowed-use records are in `ARTIFACT_MANIFEST.json`; numeric tables are the three CSV files and `nullspaces.json`. The two optical paragraphs here are an integration-ready delta; the current P6 paper source itself has not been edited by this assigned writer.